

Mock Final Exam Solutions

CPSC 406 — Algorithm Analysis
 Chapman University · Spring 2026
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1. (5 points) Turing Machines

The machine accepts exactly the language $\{a^n b^n : n \geq 1\}$.

Word	Answer
<i>aabb</i>	accept
<i>abb</i>	reject
<i>abab</i>	reject
<i>ab</i>	accept
<i>aaabbb</i>	accept

2. (5 points) Landau Notation

For $g(n) = 4n^5 + 9n^2 \log n + 7\sqrt{n}$, the dominant term is n^5 . Thus $g(n) \in \Theta(n^5)$.

Class	Answer
$\mathcal{O}(n^4)$	no
$\mathcal{O}(n^5)$	yes
$\mathcal{O}(n^6)$	yes
$\mathcal{O}(2^{\sqrt{n}})$	yes
$\mathcal{O}(n^5 \log n)$	yes

3. (5 points) CNF and Satisfiability

(a) (2 points)

$$\begin{aligned}\varphi &= (p \rightarrow (q \vee r)) \wedge \neg(q \wedge r) \\ &= (\neg p \vee q \vee r) \wedge (\neg q \vee \neg r).\end{aligned}$$

This is in CNF.

(b) (3 points) The formula is satisfiable. One satisfying assignment is

$$p = \text{false}, \quad q = \text{true}, \quad r = \text{false}.$$

4. (5 points) Logical Modeling

(a) (2 points) Let

M : the printer is in maintenance mode,
 O : the printer is online,
 H : the printer has paper,
 A : an administrator override is active,
 P : printing is allowed.

The policy and log can be formalized as

$$(M \rightarrow \neg P) \wedge ((O \wedge H) \rightarrow P) \wedge (A \rightarrow P) \wedge M \wedge O \wedge H \wedge \neg A \wedge P.$$

(b) (1 point) A corresponding set of clauses is

$$\{\{\neg M, \neg P\}, \{\neg O, \neg H, P\}, \{\neg A, P\}, \{M\}, \{O\}, \{H\}, \{\neg A\}, \{P\}\}.$$

(c) (1 point) Resolution derives the empty clause:

$$\{\neg M, \neg P\}, \{M\} \vdash \{\neg P\}, \quad \{\neg P\}, \{P\} \vdash \square.$$

Hence the log is inconsistent with the policy.

(d) (1 point) The conflict is that maintenance mode requires printing not to be allowed, but the log says printing was allowed.

5. (5 points) **Induction and Big-O Bounds**

We prove $T(n) \leq n \log_2 n + n$ by induction on powers of two.

Base case: for $n = 1$,

$$T(1) = 1 \leq 1 \log_2 1 + 1 = 1.$$

Induction step: assume $T(n/2) \leq \frac{n}{2} \log_2 \frac{n}{2} + \frac{n}{2}$ for some power of two $n \geq 2$. Then

$$\begin{aligned} T(n) &\leq 2T\left(\frac{n}{2}\right) + n \\ &\leq 2\left(\frac{n}{2} \log_2 \frac{n}{2} + \frac{n}{2}\right) + n \\ &= n \log_2 \frac{n}{2} + n + n \\ &= n(\log_2 n - 1) + 2n \\ &= n \log_2 n + n, \end{aligned}$$

Therefore $T(n) \leq n \log_2 n + n$ for all powers of two $n \geq 1$, and hence $T(n) \in \mathcal{O}(n \log n)$.

6. (5 points) **Recurrence Trees**

(a) (2 points) The tree corresponds to

$$T(n) = 3T\left(\frac{n}{4}\right) + cn,$$

with a constant-size base case, for example $T(1) = d$.

(b) (3 points) By the Master theorem, $a = 3$, $b = 4$, and $n^{\log_b a} = n^{\log_4 3}$. Since

$$cn \in \Omega(n^{\log_4 3 + \varepsilon})$$

for some $\varepsilon > 0$, and the regularity condition holds,

$$T(n) \in \mathcal{O}(n).$$

Alternatively, you can argue directly from the recurrence tree. At level i , there are 3^i subproblems, each of size $n/4^i$. Therefore the total nonrecursive work at level i is

$$3^i \cdot c \left(\frac{n}{4^i}\right) = cn \left(\frac{3}{4}\right)^i.$$

The tree has height $\log_4 n$, so the total nonrecursive work is

$$\begin{aligned} \sum_{i=0}^{\log_4 n - 1} cn \left(\frac{3}{4}\right)^i &\leq cn \sum_{i=0}^{\infty} \left(\frac{3}{4}\right)^i \\ &= cn \cdot \frac{1}{1 - \frac{3}{4}} \\ &= 4cn. \end{aligned}$$

The leaves contribute $d \cdot 3^{\log_4 n} = d n^{\log_4 3}$, which is $\mathcal{O}(n)$ because $\log_4 3 < 1$. Hence the total work is

$$\mathcal{O}(n) + \mathcal{O}(n^{\log_4 3}) = \mathcal{O}(n).$$

7. (5 points) **Random Permutations and Indicator Variables**

For each index $1 \leq i \leq n$, define the indicator random variable

$$X_i = \begin{cases} 1, & \text{if } A[i] \leq k, \\ 0, & \text{otherwise.} \end{cases}$$

Then the total number of positions containing one of the values $1, 2, \dots, k$ is

$$X = \sum_{i=1}^n X_i.$$

For any fixed index i , each value in $\{1, \dots, n\}$ is equally likely to appear in position i . Hence

$$\mathbb{P}(A[i] \leq k) = \frac{k}{n}, \quad \mathbb{E}[X_i] = \frac{k}{n}.$$

By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^n \frac{k}{n} = k.$$

8. (5 points) **Probabilistic Algorithm Analysis**

- (a) (2 points) The number of loop iterations is

$$\sum_{i=1}^n i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Therefore the worst-case running time is $\mathcal{O}(n^2)$.

- (b) (3 points) Let X be the returned value. There is one die roll per loop iteration, and each roll contributes 1 with probability $1/6$. By linearity of expectation,

$$\mathbb{E}[X] = \frac{1}{6} \cdot \frac{n(n+1)}{2} = \frac{n(n+1)}{12}.$$